Physics-Guided Prediction and Simulation-Based Control of Crane Payload Sway under Cable-Length Shift

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Abstract

This research draft investigates cable-length shift in crane payload-sway forecasting using an existing experimental dataset, with a separate simulation-based control study. A causal Python benchmark is developed from the author's previous MATLAB study. It distinguishes the forecasting information set, scenario-level validation, and retrospective evaluation. An initial extrapolation experiment compares eleven predictors on four held-out trolley-motion records. At a 1.0 s horizon, a train-calibrated oscillator achieves a mean scenario MAE of 0.758 degrees, compared with 0.907 degrees for an exploratory sensitivity-projected neural residual. The residual improves the 0.1 s error from 0.300 to 0.219 degrees but does not establish uniform superiority. An additional benchmark covers 38 scenario–axis folds in three motion families. A separate 16-context simulation pilot compares nominal and geometrically calibrated MPC under synthetic plant mismatch. These controller experiments assume state feedback and do not validate the neural predictor inside MPC. The current evidence establishes a reproducible research baseline and identifies the need for stronger adaptive-physics comparisons before a novel-method claim. An additional command-conditioned passive-memory model is implemented with a discrete storage balance. Its short-horizon gains do not uniformly exceed a same-information GP residual reference. No new physical experiments or industrial safety claims are made.

Keywords: crane sway; physics-guided forecasting; cable-length shift; residual learning; model predictive control.

1 Introduction

Predicting suspended-load motion is useful only when the available measurements, future inputs, and intended decisions are defined precisely. A predictor driven by recent sway angles can forecast approximately free motion, but cannot determine an arbitrary future operator command. Conversely, predictive control requires a model of how candidate actions change the system. These are related but distinct identification problems.

The present work extends the author's previous analysis of payload-sway prediction horizons and cable-length generalisation. That analysis used tree-based regression and neural networks on an experimental tower-crane dataset. Its subsequent audit identified the importance of physical baselines, consistent horizon indexing, and scenario-level evaluation. The current Python implementation preserves the historical results as provenance while refitting new models under an explicit protocol.

The central development question is whether a model can retain useful physical structure when transferred to a new suspension configuration. Rather than assuming that a learned correction is beneficial, we test whether frequency calibration already explains most of the improvement. The intended longer-term contribution is a data-efficient adaptation method connecting action-conditioned

prediction and control. That contribution has not yet been established by the initial results reported here. This draft records both positive and negative findings.

2 Related work and scope of the proposed contribution

Crane identification already includes multi-step symbolic predictors conditioned on cable length and payload mass[1], regularised genetic-programming identification[2], and learned NARX models embedded in nonlinear MPC[3]. Physics-informed sway prediction and simulation-to-experiment transfer have been studied for travelling mobile cranes[4]. Mao et al. combine a physical model, a Gaussian-process residual and a disturbance observer in experimentally evaluated crane MPC[5]. Therefore, combining physics, a learned correction and MPC is not a sufficient novelty claim.

Dimensionless control policies provide a framework for transfer between dynamically similar systems[6]. Orthogonal Gaussian processes and projection-regularised physics-guided neural models address the confounding of physical parameters and flexible residuals [7, 8]. The exploratory model below applies a related separation at the level of finite-horizon oscillator predictions. The specific use of trajectory sensitivities, causal calibration budgets and control-relevant evaluation remains a research candidate requiring fuller comparison with these predecessors.

The current manuscript is a development report rather than a claim that this combination constitutes a new general theory. In particular, the available physical records support retrospective forecasting. They do not contain the response to newly selected controller actions.

3 Dataset and evaluation contract

3.1 Experimental records

The public Inteco tower-crane dataset, Mendeley version 2, contains 33 operating scenarios with 9901 observations each over 99 s, sampled at $\Delta t = 0.01$ s[9, 10]. It records trolley motion, jib rotation, hoisting, and combined motions. The nominal payload is 300 g; additional records vary the payload and suspension length. Encoder resolution is approximately 0.0879 degrees. Raw XLSX files are imported without modification and identified by SHA-256 checksums. All 33 checksums match the retained historical inventory.

The first extrapolation experiment uses trolley scenarios 3, 5 and 6 for training, scenario 4 for validation, and scenarios 1, 2, 7 and 8 for evaluation. This tests both shorter and longer lengths than those seen in training. The records have already informed earlier research; the evaluation is retrospective, not a previously untouched test.

3.2 Forecast origin and information

For prediction origin k, a history of m measured angles is

$$\mathbf{h}_k = [\theta_k, \theta_{k-1}, \ldots, \theta_{k-m+1}], \qquad \mathbf{y}_k = [\theta_{k+H_1}, \ldots, \theta_{k+H_q}]. \tag{1}$$

Thus horizon H_j corresponds to exactly $H_j \Delta t$ after the newest available angle. The neural pilot uses $m = 41$ and horizons $\{0.10, 0.20, 0.35, 0.50, 0.75, 1.00, 1.50\}$ s. Prediction origins are evaluated every two samples. All input scaling and physical calibration use the training records only; early stopping uses the separate validation scenario. Overlapping windows are not independent trials.

No future actual trolley motion is supplied to the forecasting models. An active/quiet diagnostic uses backward differences of measured crane motion at the prediction origin. This label does not

imply that the entire future horizon is unforced. A fixed split at 15 s is not used to declare all later motion free decay.

4 Physical and learned forecasting models

4.1 Physical calibration and causal state fitting

A descriptive effective-length relationship is fitted inside each training fold:

$$\ell_{\text{eff}}(L) = aL + b, \qquad \omega(L) = \sqrt{g/\ell_{\text{eff}}(L)}. \tag{2}$$

The calibration uses frequency estimates from training intervals between 35 and 90 s, with a positive effective-length constraint. These intervals are identification tails, not automatically assumed unforced. The fitted offset is a model-discrepancy parameter and is not, without separate geometric measurement, a measured hook or cable offset.

At every prediction origin, oscillator coefficients are estimated from the declared history only:

$$(\hat{A}_k, \hat{B}_k) = \arg\min_{A,B} \sum_{i=0}^{m-1} \left[\theta_{k-i} - A\cos(\omega i \Delta t) + B\sin(\omega i \Delta t)\right]^2. \tag{3}$$

The future prediction is

$$\hat{\theta}^{\text{phys}}_{k+H} = \hat{A}_k \cos(\omega H \Delta t) + \hat{B}_k \sin(\omega H \Delta t). \tag{4}$$

The software also supports a damped basis; the initial benchmark uses zero damping so that the role of frequency calibration remains visible. This is a harmonic forecasting baseline, not a complete nonlinear, command-conditioned crane model.

4.2 Baseline models

The pilot compares persistence, an ideal oscillator using the recorded length directly, the calibrated oscillator, OLS, a length-varying linear model, gradient-boosted trees, an MLP ensemble, a capped Gaussian-process residual, and three phase-coordinate residual variants. The MLP and residual networks use two hidden layers with 20 units and three fixed training seeds. They are fitted jointly to the seven horizons. The Python neural and tree models are new implementations under a shared protocol, not bit-identical replicas of MATLAB training routines.

4.3 Exploratory sensitivity-projected residual

Let $\hat{\mathbf{y}}^{\text{phys}}(p)$ denote the oscillator forecast and $r_w(\mathbf{h}_k, L)$ a learned correction. Phase-coordinate features include fitted oscillator coefficients, causal history discrepancy, and the dimensionless sampling interval $\omega\Delta t$. Trajectory sensitivities to frequency and damping define $S_k = \partial\hat{\mathbf{y}}^{\text{phys}}/\partial p$. A rank-aware basis U_k spans the retained sensitivity directions. The candidate objective is

$$\mathcal{L}(w) = \mathcal{L}_{\text{forecast}}(w) + \lambda\, \mathbb{E}_k \left\| U_k^{\mathsf{T}} r_w \right\|^2. \tag{5}$$

The pilot fixes $\lambda = 0.03$ as a development choice, and compares the same residual architecture with $\lambda = 0$. Projection acts on the complete correction including its learned mean. It defines a division of model components, not guaranteed recovery of true physical parameters. It may be harmful when genuine discrepancy aligns with the penalised directions; that possibility is part of the experiment.

5 Initial experimental forecasting results

Table 1 averages MAE equally across the four evaluation scenarios. The calibrated oscillator outperforms all learned variants at 1 s in this initial split. The projected residual improves the short 0.1 s horizon but worsens the 1 s result. A uniform advantage of the candidate method is therefore not supported.

Table 1: Retrospective cable-length extrapolation MAE in degrees. Each entry is the mean of four scenario scores; neural predictions are three-seed ensembles. These scores are not independent-window inference.

Model	0.1 s	0.5 s	1.0 s
Calibrated oscillator	0.300	0.574	0.758
Phase ridge residual	0.226	0.611	0.850
Projected phase residual	0.219	0.616	0.907
Phase NN residual	0.229	0.625	0.951
GP residual	0.264	0.919	1.033
OLS	0.544	3.951	6.312
LPV	2.168	3.796	7.527
Boosted trees	1.793	4.608	7.564
Ideal oscillator	3.624	6.606	7.755
MLP ensemble	2.117	5.549	8.507
Persistence	3.286	12.772	9.380

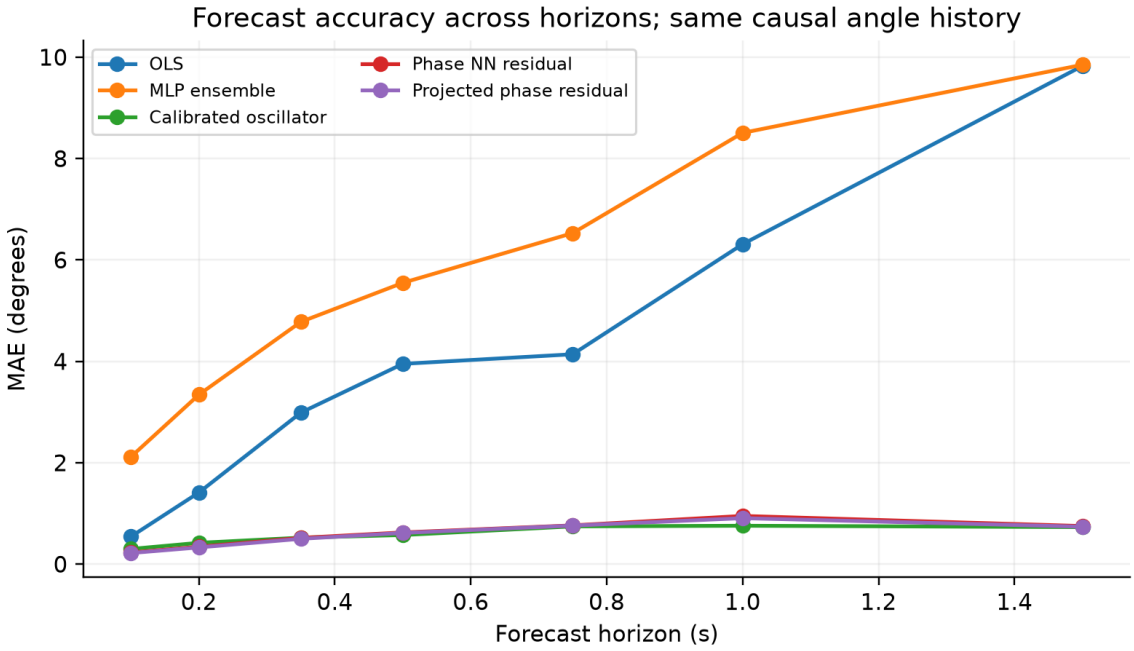


Figure 1: Horizon-dependent error in the initial extrapolation split. The plot includes unfavourable residual-learning results.

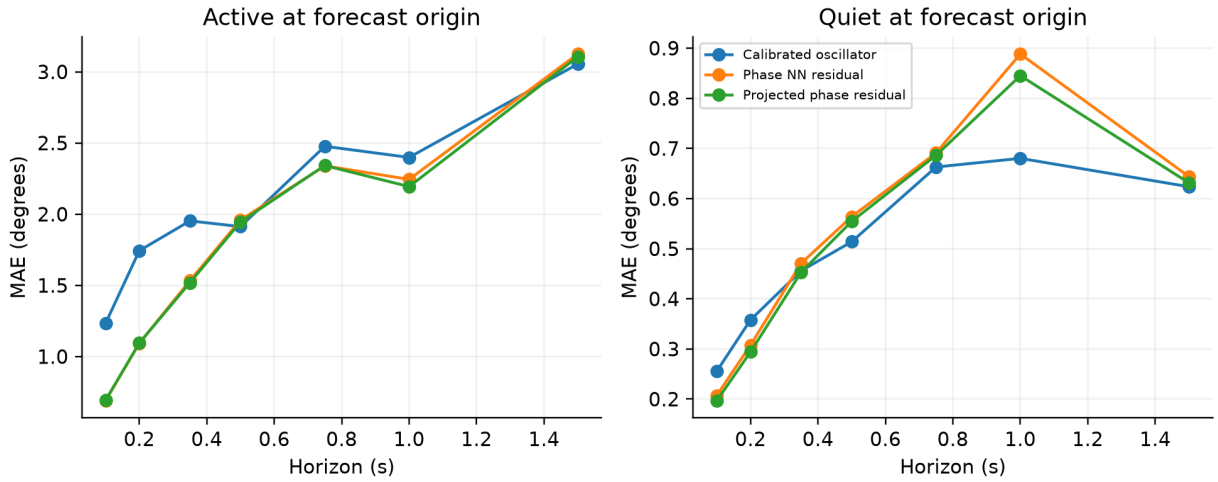


Figure 2: Scores separated by a causal activity diagnostic at the forecast origin. Quiet-at-origin does not imply an unforced future.

An additional physical-baseline benchmark evaluates all 38 scenario–axis folds of the fixed-length trolley, jib and simultaneous motion families with histories of 3 and 41 observations. Its full per-scenario results are retained in the repository. Longer history constitutes additional information and is reported separately rather than attributed to model architecture.

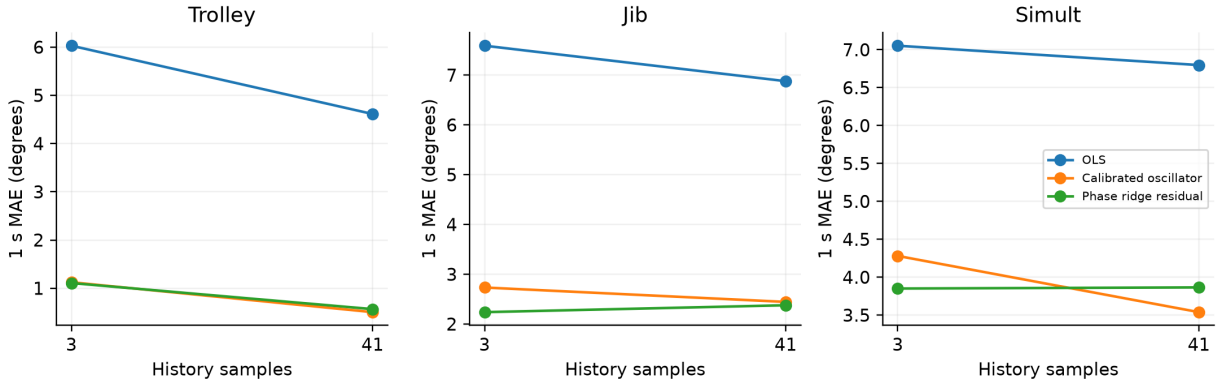


Figure 3: Initial all-family physical baselines at 1 s. Families with two angular outputs are averaged over their scenario–axis scores.

6 Action-conditioned passive-memory development

6.1 Information set and actuator identification

An additional fixed-length trolley experiment supplies a 41-sample history of sway, measured trolley position and recorded reference, together with the prescribed future reference. No future measured trolley position or acceleration enters a predictor. This information set differs from the earlier angle-only benchmark. Forecast origins are spaced by 0.1 s and rollout integration uses 0.05 s steps; scores must not be mixed with the earlier, denser origin grid.

The reference is retained in its recorded numerical scale. In training scenario 3, a reference of 0.25 coexists with a stationary measured position near 0.508 m, so treating the reference as exact

position or acceleration would be unjustified. A small recurrent actuator model is fitted to multi-step position error using scenarios 3, 5 and 6, with early stopping on scenario 4. On that validation scenario, 1 s position MAE is 2.43 mm, versus 4.86 mm for constant velocity. At 0.1 s the learned actuator is worse: 0.25 versus 0.16 mm. The actuator is frozen before sway fitting. These values do not establish an identified industrial drive model.

6.2 Phase-exact passive-memory surrogate

Output-error port-Hamiltonian identification with causal state encoding already exists[11]; so do orthogonal model augmentation [12] and power-preserving neural discretisations [13]. The present candidate is a particular fixed-length, small-angle construction, not a claim to introduce those general ideas. For $\tau = \omega t$, let $z = [q, p, r^{\mathsf{T}}]^{\mathsf{T}}$, where $p = \dot{q}/\omega$ and r contains four latent memory states. A causal encoder provides a bounded initial memory and a velocity correction. A small network supplies nonnegative damping d, positive rates Λ, and exchange coefficients c, fixed over each rollout. Its closure matrix and model storage are

$$C = \begin{bmatrix} 0 & 0 & 0 \\ 0 & -d & c^{\mathsf{T}} \\ 0 & -c & -\Lambda \end{bmatrix}, \qquad E(z) = \tfrac{1}{2}\|z\|^2. \tag{6}$$

Two Cayley closure half-steps surround an analytic forced-oscillator rotation with phase $\omega\Delta t$ and force $f = -\hat{a}_x/g$. For one closure step of dimensionless duration δ,

$$z^+ = (I - \delta C/2)^{-1}(I + \delta C/2)z, \qquad E(z^+) - E(z) = -\delta(d\bar{p}^2 + \bar{r}^{\mathsf{T}}\Lambda\bar{r}), \tag{7}$$

where $\bar{z} = (z + z^+)/2$. The central rotation supplies work $f(q^+_{\mathrm{rot}} - q^-_{\mathrm{rot}})$. Summing these terms yields an exact discrete storage balance in exact arithmetic. With no input, the full storage cannot increase. When closure is disabled, the base oscillator has no numerical phase error. These properties do not eliminate model error or imply closed-loop stability. Nonzero memory can change the modes of the complete model, and observer resets can change storage.

Independent tests check analytic rotation, storage, finite-difference gradients, batch independence, and second-order convergence to a separate adaptive solver for constant coefficients. Maximum storage-balance residual in the float32 forecasting runs is about 8.4×10^{-8} in normalised storage units. A comparison with the same encoder and no memory isolates part of the benefit of latent states; its parameter counts are 7266 without memory and 7662 with memory.

6.3 Same-information references and retained result

The neural actuator is shared across all action-conditioned methods. Three fixed seeds are averaged for each neural variant. Direct MLP and GP residual references receive the same available past packet and future command, including the deterministic actuator rollout. The GP uses up to 512 training points; four fixed kernel length scales are compared on scenario 4 before evaluation. The selected GP and MLP are forecasting references, not recursive plant models inside MPC.

Table 2: Additional retrospective trolley pilot: mean MAE in degrees across scenarios 1, 2, 7 and 8. The angle-only row has less information; the remaining rows share prescribed future commands and past trolley measurements. The origin grid differs from Table 1.

Model	0.1 s	0.5 s	1.0 s
Angle-only harmonic	0.301	0.577	0.759
Action-conditioned physics	0.245	0.546	0.757
Neural observer, no memory	0.235	0.549	0.767
PEPM, four memory states	0.228	0.549	0.765
GP action residual	0.216	0.501	0.785
MLP action residual	0.233	0.529	0.756

PEPM improves 0.1 s MAE over action-conditioned physics from 0.245 to 0.228 degrees; the same observer without memory gives 0.235 degrees. However, the GP reference achieves 0.216 degrees and is stronger at several intermediate horizons. At 1 s, PEPM gives 0.765 degrees versus 0.757 for action-conditioned physics. Thus the passive construction is implemented and numerically checked, but broad predictive superiority is not established. No thresholds or architecture settings were retuned from these evaluation scores. The proposed neural model has not yet been evaluated inside MPC.

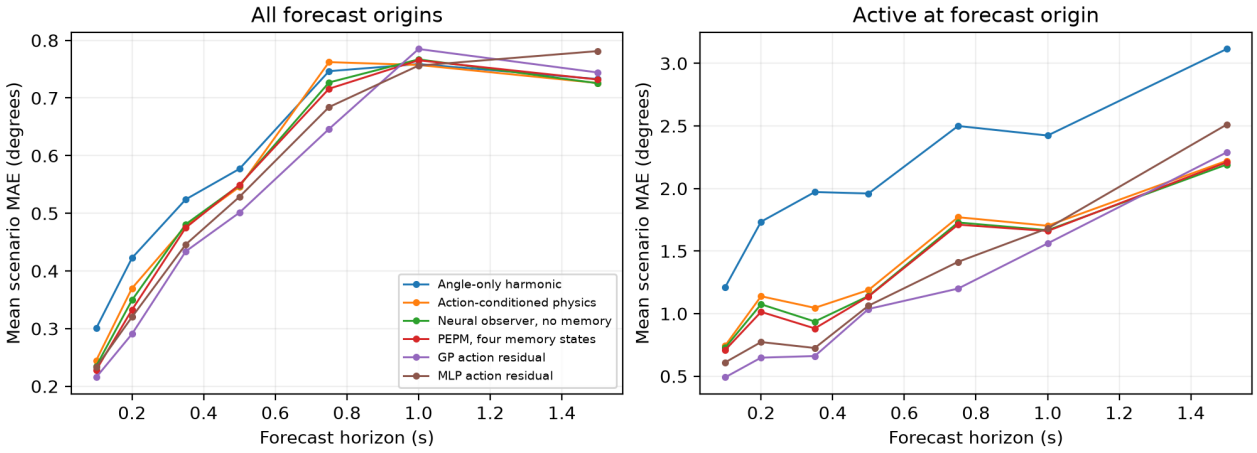


Figure 4: The action-conditioned pilot retains GP and simple-physics comparators. Active-at-origin labels use a backward-only velocity fit. Prescribed future commands change the information set relative to the angle-only harmonic reference.

7 Separate simulation-based control pilot

7.1 Plant and controller

The simulation uses a nonlinear prescribed-acceleration pendulum with actuator lag, dead zone and gain mismatch. For state $[x, v, \theta, \dot{\theta}, a_x]$, the angular equation is

$$\ddot{\theta} = -\frac{g \sin\theta + a_x \cos\theta}{\ell} - d\dot{\theta} + \eta(t). \tag{8}$$

A fourth-order Runge–Kutta plant integrator is checked against an independent adaptive solver. The MPC uses a linearised discrete-time model with a 40-step horizon at 20 Hz and a bounded

acceleration command. Identical cost weights and limits are used in both controllers. Only their effective-length models differ.

Sixteen paired synthetic contexts vary length, actuator lag, gain, damping and dead zone. Each controller receives the same initial state and physical-time disturbance sequence. The true effective length is sampled independently of the fitted calibration coefficients, within the declared synthetic mismatch range. This is not an empirically identified distribution of industrial cranes.

Exact state feedback is assumed in this first control experiment. Only input bounds are enforced by the optimiser; monitored state limits are not certified hard constraints. Solver failures trigger a predefined feedback fallback and remain part of the reporting.

7.2 Initial control results and interpretation

Table 3: Simulation-only pilot: means across 16 paired contexts. The neural forecaster is not used inside these controllers. Solve-time entries average each episode's p99; they are not a pooled p99 guarantee.

Controller	Residual RMS (deg)	Sway integral ($deg^2 s$)	Settle time (s)	p99 solve (ms)
nominal mpc	0.081	27.488	2.994	1.788
calibrated mpc	0.064	21.863	2.978	1.454

Mean residual RMS decreases from 0.0805 to 0.0639 degrees in the chosen synthetic campaign, while mean settling times are approximately 2.99 and 2.98 s. The paired mean difference is -0.0166 degrees, with a scenario-bootstrap interval of approximately $[-0.0322, -0.0037]$ degrees. This interval describes the sampled synthetic contexts, not hardware uncertainty or real-world deployment. All 32 pilot episodes settled under the declared criterion, with no reported solver failures or 50 ms solve-budget overruns. Timing covers the optimiser call, not sensor acquisition or a real operating system's end-to-end control latency.

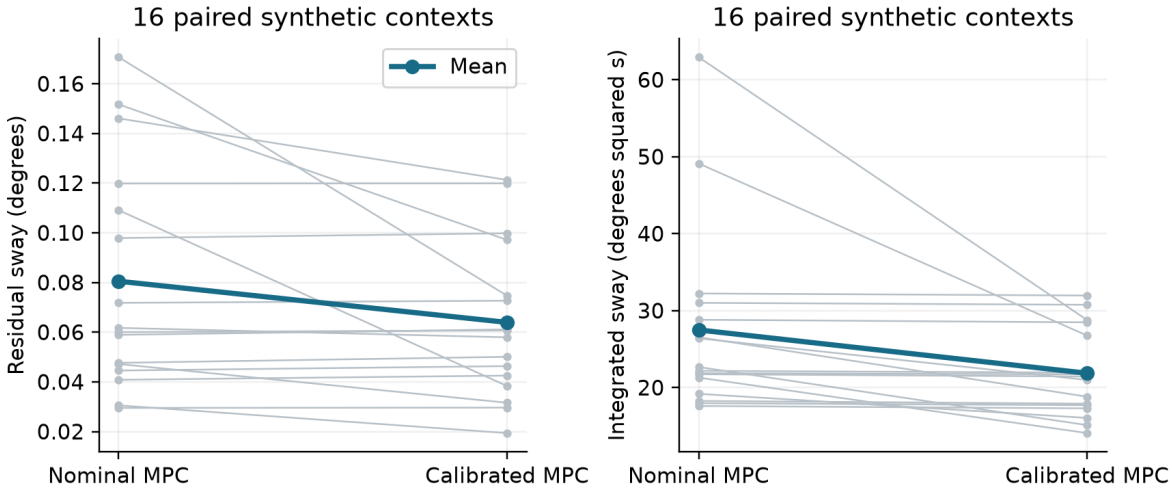


Figure 5: Paired control metrics in the initial synthetic campaign. Thin lines retain every context; the thick line shows the mean.

This pilot demonstrates that the implemented control pipeline can test model mismatch. It does

not establish that the projected neural predictor improves MPC, nor replace an adaptive-physics control baseline.

8 Limitations and remaining research

The all-family neural benchmark also completes 38 scenario–axis folds, with three seeds for each of the three neural variants (342 network fits). Table 4 shows that the ranking depends on the motion axis: learned corrections help some transverse outputs, while calibrated physics remains strong on longitudinal motion. GP residuals are an especially important comparator and are not consistently surpassed by the projected network.

Table 4: Mean 1 s MAE in degrees in the retrospective all-family neural benchmark, with 41-sample histories and fixed development settings.

Family	Axis	Calibrated	GP residual	Projected NN
jib	beta	2.989	2.412	2.500
jib	theta	1.914	2.290	2.074
simult	beta	3.065	2.237	2.375
simult	theta	4.023	4.951	4.229
trolley	theta	0.502	0.904	0.534

A separate adaptive-physics reference identifies frequency, damping and an input coefficient from a causal prefix, using measured trolley motion inside that prefix. Budgets of 0, 1, 3, 5 and 10 prior oscillator periods are scored on the same later interval beginning at 25 s. All nonzero budgets worsen the mean 1 s error in this initial experiment: 0.673 degrees without adaptation versus 1.099 degrees at ten periods. This comparison has a different calibration information set from the angle-only zero-shot benchmark. It shows that the chosen local estimator is inadequate, not that all forms of online adaptation fail.

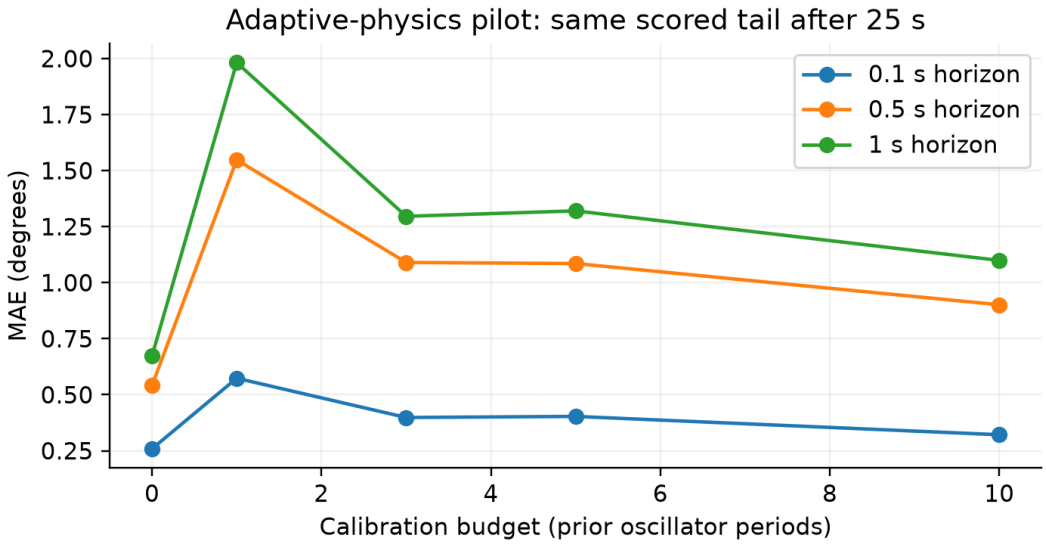


Figure 6: Negative adaptation result retained as a reference. All budgets share the same scored tail; no calibration-prefix target enters evaluation.

An additional, fixed development protocol fits frequency-only and damped candidates on the first 60% of each available prefix and evaluates them on the last 40%. Acceptance requires at least two non-overlapping one-second target blocks, improvement in mean MAE of at least 5%, no worse MAE at any of the three horizons, and wins on at least 75% of those blocks. Accepted candidates would be retained without refitting. These thresholds were fixed before this run and were not tuned from its results. The same protocol first processes development scenario 4, then the four evaluation scenarios. It uses angles only, whereas the earlier ungated estimator also uses measured trolley motion during calibration. No configuration update is accepted in this pilot. Across the 16 nonzero-budget evaluation cases, nine lack sufficient prefix data and seven fail the validation criteria. Consequently, the 1 s score remains 0.673 degrees at every budget. This is abstention, not successful adaptation or evidence of a guarantee against future degradation.

The main methodological hypothesis remains unconfirmed. Required next experiments include adaptive physical models under equal causal calibration budgets, command-conditioned dynamics with an identified input path, broader neural scenario folds, and independent simulation objects. Payload-shape and hoisting records are imported but do not yet provide a completed generalisation result. A learned residual cannot be credited for an advantage caused by more history or better frequency calibration.

The experimental records are few at the scenario level and contain no new independent repetitions. Some parameter effects are confounded with trajectory or payload configuration. Neither dense sampling nor multiple neural seeds removes that limitation. Simulation mismatch ranges and full-state observations further limit the control claims.

The project does not currently supply a stability proof, a distribution-free safety guarantee, a comparison against proprietary industrial anti-sway products, or evidence of full-scale transfer. Journal selection and the final novelty claim should follow the completed evidence rather than precede it.

9 Reproducibility and research provenance

The Python repository retains scenario splits, raw-data checksums, configuration hashes, package versions, forecast arrays, calibration fits, neural checkpoints, and episode traces. Initial runs predate the first repository commit; their post-run source snapshot is labelled as such. Subsequent runs additionally record source-file hashes. The historical OLS protocol is independently reproduced for all 24 retained comparisons with a maximum discrepancy below 10^{-12} degrees. That reproduction intentionally preserves the old indexing and is separate from the corrected forecast benchmark.

The study builds on Ratmir Kutyrev's earlier research and attributes the public experimental measurements to their creators. AI assistance was used in research planning, implementation, analysis scripts and manuscript preparation. The manuscript remains subject to human scientific review, verification of claims and final author approval. This paragraph must be reviewed against the chosen journal's policy before submission. No unperformed human or experimental contribution is assigned to a coauthor.

10 Conclusion of the current development stage

The first implementation establishes a causal, scenario-separated Python benchmark and a distinct simulation-based control pipeline. Calibrated physics remains a strong long-horizon reference, while learned corrections have mixed horizon-dependent effects. The additional action-conditioned passive-memory model supplies a checked discrete storage balance but does not uniformly outperform a GP

residual. Further work must resolve input-path uncertainty, test coupled dynamics and establish a predictive advantage before a state-of-the-art-method claim is supported.

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